

Multi-Task DAG (Student Marks Processing)

1. Create New DAG File

```
cat > student_marks_dag.py
```

2. DAG Code (New Logic with File Handling + XCom)

```
from airflow import DAG
from airflow.operators.python import PythonOperator
from datetime import datetime

def create_marks_file():
    with open("/tmp/student_marks.txt", "w") as f:
        f.write("Math,80\n")
        f.write("Science,75\n")
        f.write("English,90\n")
        f.write("Python,95\n")

def read_marks_file():
    with open("/tmp/student_marks.txt", "r") as f:
        for line in f:
            print(line.strip())

def calculate_total():
    total = 0
    with open("/tmp/student_marks.txt", "r") as f:
        for line in f:
            marks = int(line.strip().split(",")[1])
            total += marks
    print(f"Total Marks = {total}")
    return total

def percentage_calculation(ti):
    total = ti.xcom_pull(task_ids="calculate_total")
    percentage = total / 4
    print(f"Percentage = {percentage}")
    return percentage

def generate_result(ti):
    total = ti.xcom_pull(task_ids="calculate_total")
```

```

        with open("/tmp/result.txt", "w") as f:
            f.write("Student Result Summary\n")
            f.write(f"Total Marks = {total}\n")
            f.write("Result = PASS\n")

with DAG(
    dag_id="student_marks_dag",
    start_date=datetime(2025, 1, 1),
    schedule="@daily",
    catchup=False
) as dag:

    t1 = PythonOperator(task_id="create_marks_file",
python_callable=create_marks_file)
    t2 = PythonOperator(task_id="read_marks_file",
python_callable=read_marks_file)
    t3 = PythonOperator(task_id="calculate_total",
python_callable=calculate_total)
    t4 = PythonOperator(task_id="percentage_calculation",
python_callable=percentage_calculation)
    t5 = PythonOperator(task_id="generate_result",
python_callable=generate_result)

    t1 >> t2 >> t3 >> t4 >> t5

```

Enable and Trigger the DAG

1. Search for:

student_marks_dag

2. Turn ON the toggle
3. Click  **Trigger DAG**

Verify Generated Files

- Go back to terminal:
docker exec -it airflow bash
- Check marks file:
cat /tmp/student_marks.txt
- Check result file:
cat /tmp/result.txt

Dag: student_daily_task_dag
 Dag Run: manual_2026-05-07T10:03:24.412569+00:00

Logical Date: 2026-05-07 15:33:35
 Run Type: Manual
 Start Date: 2026-05-07 15:33:25
 End Date: 2026-05-07 15:33:39
 Duration: 00:00:14.234
 Triggering User: admin
 Dag Version: v1

Task Instances: Asset Events: Audit Log: Code: Details

+ Filter

4 Task Instances

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration	
sleep_task	sleep_task	Success	2026-05-07 15:33:38	1	PythonOperator	00:00:00.079	✓ / ✕
complete_assignment_task		Success	2026-05-07 15:33:37	1	PythonOperator	00:00:00.131	✓ / ✕
attend_class_task		Success	2026-05-07 15:33:36	1	PythonOperator	00:00:00.100	✓ / ✕
wake_up_task		Success	2026-05-07 15:33:35	1	PythonOperator	00:00:00.109	✓ / ✕

```

root@ubuntu:~$ docker exec -it airflow bash
airflow@ab161ec4145e:/opt/airflow$ cd /opt/airflow/dags
airflow@ab161ec4145e:/opt/airflow/dags$ ls
__pycache__  first_dag.py  student_dag.py
airflow@ab161ec4145e:/opt/airflow/dags$ cat /tmp/student_marks.txt
Math,80
Science,75
English,90
Python,95
airflow@ab161ec4145e:/opt/airflow/dags$ cat /tmp/result.txt
Student Result Summary
Total Marks = 340
Result = PASS
airflow@ab161ec4145e:/opt/airflow/dags$

```